Capgemini 2027 — AI Literacy + Technical Assessment

200 Hard Scenario-Based Practice MCQs

AI foundations, prompting, advanced AI systems, responsible AI, programming, DSA, software engineering and modern engineering awareness

AI Foundations & Generative AI

1. A support team compares a traditional classifier with a foundation model. The latter can be adapted to summarization, coding and classification with task-specific prompting. Which property best explains this flexibility?

A. It is trained only on one fixed label set.
B. It learns broad representations that can support many downstream tasks.
C. It stores every possible answer as a database row.
D. It executes only manually written business rules.

2. A language model receives the same context but produces different plausible continuations under different sampling settings. What is changing at generation time?

A. The model's training dataset is being rewritten.
B. The probability distribution used for token selection is being sampled differently.
C. The tokenizer is permanently learning new words.
D. The model is converting from supervised to unsupervised learning.

3. An engineer asks a model for a newly issued company policy, but the model has no retrieval source and predates the policy. What is the primary risk?

A. The model cannot generate any text about policies.
B. The model may produce an answer unsupported by the newer information.
C. The model will automatically query the company's database.
D. The model must return an empty string for every unknown fact.

4. A team needs to turn a paragraph into a concise set of bullet points while preserving the important facts. Which task is most directly involved?

A. Summarization
B. Classification
C. Tokenization
D. Embedding generation

5. A model converts the phrase 'payment failed' into a numerical vector used to find semantically similar phrases. What is that representation commonly called?

A. Embedding
B. Checksum
C. Parse tree
D. Session token

6. A developer notices that two semantically similar sentences receive nearby vectors from an embedding model. What capability does this enable?

A. Semantic similarity search
B. Guaranteed factual verification
C. Lossless compression
D. Source-code compilation

7. A text-generation system predicts one token at a time conditioned on previous context. Which statement is most accurate?

A. It retrieves a complete answer for every sentence.

B. It generates a sequence by estimating likely next tokens.

C. It follows a fixed grammar that guarantees truth.

D. It searches the web before producing every token.

8. A team increases the temperature of a generative model and observes more varied responses. What trade-off are they exploring?

A. Diversity versus predictability

B. Storage versus network bandwidth

C. Encryption versus authentication

D. Indexing versus normalization

9. A model produces a fluent explanation containing a nonexistent API. Which conclusion is safest?

A. Fluency proves the API exists.

B. Generated content should be verified before being trusted as factual.

C. Increasing the vocabulary guarantees correctness.

D. The model has necessarily accessed a hidden API registry.

10. A text model has a maximum context window. A developer keeps appending previous conversation turns until earlier instructions disappear from effective context. What constraint is being observed?

A. Context-window limitation

B. Database normalization

C. CPU cache invalidation

D. Network packet fragmentation

11. A tokenizer splits an unfamiliar word into several smaller units instead of treating it as one vocabulary item. Why can this happen?

A. Tokenizers can represent text using subword units.

B. Every word must correspond to exactly one token.

C. Tokenization requires an internet connection.

D. Tokens are always equal to characters.

12. A model is pretrained on a broad corpus and later adapted to a narrower business task using additional training. What does the second stage represent?

A. Fine-tuning

B. DNS resolution

C. Static linking

D. Data deduplication

13. A team evaluates a classifier using precision and recall because the positive class is rare. Why might accuracy alone be misleading?

A. Accuracy can hide poor performance on an important minority class.

B. Accuracy cannot be calculated for classification.

C. Precision and recall are only network metrics.

D. Accuracy automatically ignores false positives.

14. A fraud model flags 95% of transactions correctly overall, but misses many actual fraud cases. Which metric should the team inspect closely?

A. Recall

B. Token count

C. Latency only

D. Database cardinality

15. A model assigns probabilities to several possible classes, and the probabilities sum to one. What is the model expressing?

A. A normalized distribution over candidate classes

B. A guaranteed ranking of real-world truth

C. A database transaction

D. A deterministic rule table

16. A generative model is asked to continue a sentence and chooses among several plausible tokens. Why can two runs differ even with identical input?

A. Sampling can introduce stochasticity.

B. The tokenizer changes the input every run.

C. The training corpus is modified during inference.

D. The CPU changes the model's learned weights.

17. A team compares a small specialized model with a much larger general model for a fixed classification task. Which consideration is most engineering-relevant?

A. Accuracy, latency, cost and task fit should be evaluated together.

B. The largest model is always optimal.

C. The smallest model is always more accurate.

D. Model size alone determines production suitability.

18. An LLM receives a long document and is asked a question whose answer appears near the end. Which resource most directly limits how much surrounding text can be processed at once?

A. Context window

B. TCP receive window

C. Disk partition table

D. CPU instruction set

19. A model can produce code but sometimes uses a library function that does not exist. Which capability is missing if the system relies only on generation?

A. Reliable external verification of generated claims

B. Tokenization

C. Text generation

D. Language modeling

20. A team wants to compare two model versions using the same fixed test set and metric definitions. What practice are they implementing?

A. Controlled evaluation

B. Random deployment

C. Data encryption

D. Schema migration

21. A classifier's decision threshold is raised so fewer examples are labeled positive. What typically happens to the trade-off?

A. Precision may increase while recall may decrease.

B. Recall must always increase.

C. Both metrics become exactly one.

D. The model stops producing probabilities.

22. A model is trained to predict masked words from surrounding text rather than directly generate a response. Which learning setup does this resemble?

A. Self-supervised language modeling

B. Manual rule authoring

C. Database indexing

D. Packet routing

23. A developer asks a model to output a 300-word answer but receives a much shorter answer because a generation limit was configured. What parameter family is relevant?

A. Output-token limit

B. DNS TTL

C. SQL isolation level

D. CPU clock multiplier

24. A team observes that increasing model size improves benchmark performance but also increases inference cost. What engineering lesson follows?

A. Capability improvements can introduce latency and cost trade-offs.

B. Larger models have zero operational trade-offs.

C. Model size determines database consistency.

D. Cost is independent of inference workload.

25. A system needs reliable answers about a private document collection that was updated yesterday. Which architecture is more appropriate than relying only on pretrained memory?

A. Retrieval-augmented generation using current documents

B. Increasing temperature

C. Removing the document collection

D. Replacing the model with a tokenizer.

Prompt Engineering & AI Productivity

26. A developer gives an AI assistant the goal, relevant repository context, constraints, examples and required output format. Why is this stronger than 'fix this code'?

A. It reduces ambiguity and defines an actionable task.
B. It permanently retrains the model.
C. It guarantees the code will compile.
D. It removes the need for testing.

27. A prompt contains three input-output examples before asking the model to classify a fourth input. Which technique is this?

A. Few-shot prompting
B. Zero-shot prompting
C. Fine-tuning
D. Retrieval indexing

28. A model receives an instruction without examples and must perform the task from the instruction alone. What is this commonly called?

A. Zero-shot prompting
B. Few-shot prompting
C. Parameter training
D. Distillation

29. An application parses model output programmatically. Which instruction is most useful?

A. Return only valid JSON with the specified fields and no surrounding prose.
B. Answer creatively in any format.
C. Use as many explanations as possible.
D. Return whichever structure seems convenient.

30. An AI coding assistant proposes a refactor. The developer requires unchanged public behavior and tests for edge cases. What role do those constraints play?

A. They define acceptance criteria.
B. They guarantee semantic equivalence automatically.
C. They replace the test suite.
D. They prevent all implementation changes.

31. A prompt asks an AI to optimize a database query but omits schema, indexes and workload. What should be added first?

A. Relevant schema, indexes, workload and correctness constraints
B. A higher temperature
C. A request to guess missing details
D. More decorative wording

32. A developer asks an AI assistant to review code for security vulnerabilities and return severity, evidence and remediation in a fixed schema. What benefit does the schema provide?

A. More predictable downstream processing
B. Permanent model retraining
C. Guaranteed vulnerability detection
D. Elimination of false positives

33. A prompt says 'be accurate' but does not define what counts as correct. Which improvement is strongest?

A. Provide measurable criteria or examples of acceptable output.
B. Remove all constraints.
C. Ask the model to invent the evaluation criteria.
D. Increase temperature.

34. An analyst asks an AI system to compare two designs using cost, latency and reliability, then rank them with justification. What makes the prompt stronger?

A. It specifies evaluation dimensions and the expected decision.

B. It leaves the decision criteria implicit.

C. It forbids comparison.

D. It asks only for a one-word answer.

35. A team uses the same prompt template for hundreds of tickets and changes only the ticket-specific fields. What advantage does templating provide?

A. Consistency and easier maintenance

B. Guaranteed correctness

C. Automatic fine-tuning

D. Elimination of input validation

36. A prompt contains conflicting instructions: one says 'return JSON only' and another says 'explain your reasoning in prose.' What is the main engineering problem?

A. Ambiguous or conflicting requirements

B. Excessive database normalization

C. Insufficient CPU cores

D. Lack of DNS caching

37. An AI assistant is asked to modify production configuration directly after generating a proposed change. What is the safer workflow?

A. Review, validate and test the proposed change before execution.

B. Execute immediately because generated configuration is self-validating.

C. Increase temperature before deployment.

D. Disable logging to reduce noise.

38. A developer wants the model to preserve an API contract while changing internal implementation. Which prompt detail is most important?

A. Explicitly state the public contract and prohibited behavior changes.

B. Ask for any implementation that looks cleaner.

C. Remove all examples.

D. Request maximum creativity.

39. An AI assistant repeatedly invents values when information is missing. Which instruction can reduce this behavior?

A. State that unknown information must be marked unknown rather than guessed.

B. Require a confident answer regardless of evidence.

C. Increase temperature.

D. Remove source context.

40. A team wants to improve a prompt systematically rather than by intuition. What should they do first?

A. Define representative test cases and evaluation criteria.

B. Change several prompt variables simultaneously without measuring.

C. Judge only the most impressive output.

D. Remove difficult examples.

41. A prompt includes a long irrelevant policy document even though only two clauses matter. What is the likely issue?

A. Unnecessary context can increase distraction and token usage.

B. More context always improves accuracy.

C. The model will automatically ignore all irrelevant content.

D. Context length has no operational effect.

42. A developer asks an AI to generate unit tests but provides no expected behavior. What is the main limitation?

A. The model lacks a reliable oracle for deciding what behavior should pass.

B. Unit tests cannot be generated from code.

C. Test cases require a database.

D. The model cannot read source code.

43. A prompt asks an AI to produce SQL and explicitly forbids destructive statements. Why is this useful?

A. It states a safety constraint that can be checked before execution.

B. It guarantees the SQL is syntactically valid.

C. It eliminates the need for permissions.

D. It changes the database engine.

44. An AI-generated summary omits a critical exception from the source document. What should the team improve?

A. Evaluation should include cases that test preservation of critical details.

B. Increase temperature so summaries become longer.

C. Remove the source document.

D. Accept fluency as the quality metric.

45. A coding assistant is used to migrate an API from version 1 to version 2. Which context is most valuable?

A. The old contract, new contract, affected callers and compatibility requirements

B. Only the function name

C. A request to make arbitrary changes

D. The developer's preferred font

46. A prompt requests 'the best algorithm' without defining constraints. Why is that weak?

A. Algorithm choice depends on input size, operations, memory and correctness requirements.

B. There is always one universally best algorithm.

C. Algorithms are independent of constraints.

D. Complexity never affects engineering decisions.

47. A team wants an AI assistant to review pull requests consistently. Which approach is strongest?

A. Use a stable review checklist, repository context and explicit severity criteria.

B. Ask the model to judge each PR using unrelated criteria.

C. Change the task wording randomly each time.

D. Ignore false positives.

48. A model returns valid JSON but places an incorrect customer ID in it. What does this demonstrate?

A. Format correctness does not imply semantic correctness.

B. Valid JSON guarantees factual accuracy.

C. Schema validation replaces business validation.

D. Structured output cannot contain errors.

49. A developer asks an AI to explain a complex function to a junior engineer and specifies the function's purpose and audience. Why does audience context matter?

A. It helps calibrate terminology, depth and examples.

B. It changes the source code automatically.

C. It guarantees the explanation is correct.

D. It eliminates the need to inspect the function.

50. A team compares two prompt versions on the same held-out examples. What are they effectively performing?

A. Prompt evaluation through a controlled comparison

B. Database replication

C. Model pretraining

D. Network load balancing

Advanced AI Systems

51. A chatbot must answer questions from a frequently changing internal knowledge base. Which component most directly provides current information?

A. A retrieval layer over indexed documents
B. A higher sampling temperature
C. A larger tokenizer vocabulary
D. A static system prompt alone

52. A document-search system converts documents and queries into vectors before retrieving nearby items. What does the vector index primarily support?

A. Similarity-based retrieval
B. Password hashing
C. CPU scheduling
D. Source compilation

53. A retrieval system returns highly similar but outdated documents. Which additional signal is most useful?

A. Metadata such as document version or effective date
B. Higher generation temperature
C. More random sampling
D. Removing timestamps

54. An agent can call a payment API. Which control is most important before allowing an irreversible refund action?

A. Authorization and policy checks before tool execution
B. Increasing model temperature
C. Removing audit logs
D. Letting the model choose any account

55. A tool-using agent receives a tool result that conflicts with its initial assumption. What should a robust system do?

A. Re-evaluate the plan using the new evidence.
B. Ignore the tool result.
C. Repeat the same action indefinitely.
D. Treat the first assumption as authoritative.

56. A RAG system retrieves five documents and passes them to the model. Why is retrieval quality a critical part of answer quality?

A. The generator can only use retrieved evidence effectively if relevant information is surfaced.
B. Generation quality is independent of context.
C. Retrieval affects only UI formatting.
D. Irrelevant documents always improve grounding.

57. A company wants a model to answer only from approved internal documents and refuse unsupported questions. Which design is strongest?

A. Retrieval with source constraints plus refusal/grounding checks
B. High temperature generation
C. No document retrieval
D. Unrestricted web search

58. A vector database stores embeddings but not the original document text. What additional mechanism may be needed to construct an answer?

A. A mapping from retrieved vector records to source content
B. A CPU scheduler
C. A DNS resolver
D. A compiler linker

59. A retrieval system uses very large chunks, causing each result to contain unrelated information. What can improve retrieval precision?

A. More appropriate chunking based on document structure and semantic boundaries

B. Larger chunks without limit

C. Removing metadata

D. Randomly deleting source text

60. An agent repeatedly calls the same tool because it fails to detect that the goal is already satisfied. What system property is missing?

A. Termination or state-based stopping logic

B. Tokenization

C. Database normalization

D. Encryption

61. A production RAG service becomes slow as the document collection grows. Which optimization is most directly relevant?

A. Efficient indexing and retrieval rather than scanning every document

B. Increasing prompt verbosity

C. Increasing generation temperature

D. Removing all metadata

62. An agent has permission to read customer records but should not expose one customer's data to another. What boundary is essential?

A. Authorization enforced independently of model instructions

B. A larger context window

C. A lower temperature

D. A longer system prompt only

63. A team wants to know whether retrieved documents actually support generated claims. What evaluation approach is appropriate?

A. Measure retrieval relevance and answer faithfulness separately.

B. Measure only response length.

C. Measure only token probability.

D. Skip retrieval evaluation if responses sound fluent.

64. A multimodal model receives an image and text question and produces a textual answer. What distinguishes the workload?

A. Information from multiple input modalities is jointly processed.

B. Only tabular data can be processed.

C. The model must be a relational database.

D. The image is necessarily converted to SQL.

65. A speech application converts spoken audio into text before sending it to an LLM. Which component performs the first transformation?

A. Automatic speech recognition

B. Text-to-speech

C. Vector indexing

D. SQL parsing

66. A voice assistant generates text and then speaks the response aloud. Which component converts generated text to audio?

A. Text-to-speech

B. Speech-to-text

C. DNS

D. Embedding retrieval

67. A model is exposed through an API and requests are arriving faster than the model can process them. Which mechanism can protect the service?

A. Rate limiting or queue-based backpressure
B. Increasing prompt length
C. Removing authentication
D. Disabling monitoring

68. A team caches embeddings for documents that have not changed. What is the main benefit?

A. Avoiding repeated computation for unchanged content
B. Increasing factual uncertainty
C. Removing the need for retrieval
D. Guaranteeing semantic correctness

69. An AI application uses a model from a third party and sends customer records in prompts. What architectural concern becomes important?

A. Data handling, privacy and access boundaries
B. Only UI color selection
C. CPU instruction ordering
D. DNS TTL alone

70. An agent can access multiple tools with overlapping capabilities. What helps reduce unsafe tool selection?

A. Explicit tool descriptions, permissions and validation
B. Giving every tool unrestricted access
C. Removing tool schemas
D. Allowing arbitrary tool arguments

71. A retrieval system uses cosine similarity between normalized embedding vectors. What does a higher cosine similarity generally indicate?

A. Greater directional similarity between the vectors
B. Larger disk capacity
C. Higher network bandwidth
D. More database rows

72. A production AI service needs reproducible behavior for a compliance test. Which configuration is most relevant?

A. Controlled model/version, prompt and sampling configuration
B. Randomly changing the prompt on each request
C. Unbounded tool selection
D. Disabling version tracking

73. An AI agent's plan can contain many steps and tool calls. Why is a maximum step budget useful?

A. It limits runaway loops and bounds resource consumption.
B. It guarantees every plan is correct.
C. It removes the need for authorization.
D. It increases model knowledge.

74. A system retrieves confidential documents based on a user's query. What must be checked before returning a retrieved document?

A. Whether the requesting user is authorized to access it
B. Whether the document has the longest title
C. Whether the model temperature is low
D. Whether the document is stored alphabetically

75. A team observes that a RAG system gives excellent answers when retrieval is relevant but poor answers when retrieval misses the key document. What does this reveal?

A. Retrieval recall is an important upstream dependency for grounded generation.

B. Generation completely eliminates retrieval errors.

C. Temperature determines document availability.

D. Prompt formatting guarantees retrieval recall.

Responsible AI & Evaluation

76. A hiring model learns from historical decisions that systematically favored one group. What should the team investigate?

A. Whether historical data encodes discriminatory patterns and whether mitigation is needed
B. Whether to increase generation temperature
C. Whether to remove all evaluation metrics
D. Whether historical decisions are automatically unbiased

77. An AI assistant gives a confident medical claim with no supporting evidence. What is an appropriate production control for high-impact use?

A. Require evidence/verification and appropriate human oversight before consequential decisions.
B. Increase randomness.
C. Present the claim as authoritative.
D. Remove the audit trail.

78. A model performs well on a test set but poorly after deployment because the live population differs from the training distribution. What issue is indicated?

A. Distribution shift
B. Tokenization failure
C. DNS failure
D. SQL injection

79. A benchmark is repeatedly tuned until a model performs exceptionally well on it, but unseen examples are weak. What risk does this illustrate?

A. Overfitting to the evaluation set
B. Strong generalization
C. Perfect calibration
D. Data encryption

80. A team wants to evaluate whether a chatbot refuses unsafe requests consistently. Which test design is strongest?

A. Include diverse adversarial and boundary cases with explicit pass/fail criteria.
B. Test only friendly questions.
C. Evaluate only response length.
D. Use one example repeatedly.

81. A model's confidence score is high even when it is often wrong. What should the team evaluate?

A. Calibration between predicted confidence and observed correctness
B. Font size
C. Database indexing
D. Network topology

82. Two groups receive different error rates from a classification system. What does this suggest?

A. The team should investigate subgroup performance and potential fairness concerns.
B. The model is automatically fair because total accuracy is high.
C. Subgroup metrics are never relevant.
D. The difference proves intentional discrimination without further analysis.

83. A model is evaluated only on average accuracy, while a small but safety-critical class is frequently missed. What is a better approach?

A. Include class-specific and risk-sensitive metrics.
B. Remove the rare class from evaluation.
C. Report only overall accuracy.
D. Increase model temperature.

84. A generated answer cites a real source, but the source does not support the claim. What failure is this?

A. Unsupported or misleading attribution
B. Successful grounding
C. Perfect retrieval
D. Schema validation

85. A team logs every user prompt for debugging, including sensitive customer data. What should they consider?

A. Data minimization, access controls, retention and privacy requirements
B. Increasing log verbosity without controls
C. Publicly sharing logs
D. Disabling authentication

86. An AI system makes a loan recommendation. Which design best supports accountability?

A. Maintain traceable inputs, model/version information, decision criteria and appropriate review mechanisms.
B. Hide all decision records.
C. Allow undocumented model changes.
D. Treat every recommendation as automatically correct.

87. A fairness metric improves for one group but worsens substantially for another. What should the team do?

A. Analyze the trade-off against the actual risk and policy requirements rather than optimizing one metric blindly.
B. Declare the model fair solely because one metric improved.
C. Delete the evaluation data.
D. Increase temperature.

88. A test dataset contains examples that were also used during training. Why is this problematic?

A. It can inflate measured performance through data leakage.
B. It always reduces measured accuracy.
C. It proves generalization.
D. It is required for unbiased evaluation.

89. A team changes the model, prompt and retrieval corpus simultaneously and then observes an accuracy increase. Why is attribution difficult?

A. Multiple variables changed, so the cause of the improvement is confounded.
B. Any improvement must come from the model.
C. Retrieval cannot affect results.
D. Prompt changes are never measurable.

90. An AI feature is released to production without monitoring for error rates or drift. What engineering weakness is present?

A. Lack of post-deployment observability and evaluation
B. Excessive unit testing
C. Strong governance
D. Complete rollback readiness

91. A model's performance declines gradually as customer behavior changes. What should trigger investigation?

A. Monitoring signals indicating data or concept drift
B. Only server CPU temperature
C. Only UI response length
D. A fixed calendar date regardless of metrics

92. A model provides a recommendation that a human reviewer accepts without checking because the model sounds confident. What risk is illustrated?

A. Automation bias
B. Data normalization
C. Packet loss
D. Cache locality

93. A safety-critical AI system has an uncertain prediction. Which response is generally safer?

A. Escalate, abstain or request additional evidence when uncertainty crosses a defined threshold.
B. Force a confident answer.
C. Hide uncertainty from the operator.
D. Randomly choose a class.

94. A team wants to compare two model versions fairly. What should remain controlled?

A. Evaluation data, metric definitions and test procedure
B. Only the UI theme
C. The number of engineers observing the test
D. The order of unrelated meetings

95. A red-team exercise deliberately tries prompt injection, data exfiltration and unsafe tool use. What is its purpose?

A. Identify failure modes before or during controlled deployment.
B. Increase model size.
C. Replace all monitoring.
D. Guarantee zero future vulnerabilities.

96. A model's output is used to make a high-impact decision. Which principle is most appropriate?

A. The level of oversight and validation should reflect the potential harm of errors.
B. All decisions should be fully automated regardless of risk.
C. High-impact decisions require no logging.
D. Risk is determined only by model size.

97. A dataset contains many duplicate records from one source. What evaluation problem can duplicates create?

A. They can overweight that pattern and make the dataset less representative.
B. They always improve fairness.
C. They eliminate sampling bias.
D. They guarantee independence between examples.

98. A model is accurate overall but its performance drops sharply for a specific language. What should be added to evaluation?

A. Language- or subgroup-specific test coverage
B. Only more aggregate accuracy
C. No additional testing
D. A higher generation temperature

99. A team reports that a model is '90% accurate' without stating the dataset, metric definition or evaluation conditions. What is missing?

A. Evaluation context needed to interpret the claim
B. A larger context window
C. A database index
D. A tokenizer

100. A deployed model is updated and a previously passing safety test now fails. What should the team do?

A. Treat the regression as a release-blocking signal until investigated and addressed.

B. Ignore it because the model is newer.

C. Delete the failing test.

D. Increase randomness.

Programming Logic & Problem Solving

101. An array contains values 2, 4, 6, 8 and a loop repeatedly adds the current value to a running total. What is the final total?

A. 16
B. 18
C. 20
D. 24

102. A loop starts with i = 1 and doubles i each iteration until i exceeds n. What is its time complexity?

A. O(log n)
B. O(n)
C. O(n log n)
D. $O(n^2)$

103. A loop increments i by 3 while i < n. What is its asymptotic time complexity?

A. O(1)
B. O(log n)
C. O(n)
D. $O(n^2)$

104. A function recursively calls itself twice on input n-1 without memoization and performs constant work otherwise. What is the dominant growth pattern?

A. Exponential
B. Logarithmic
C. Linear
D. Constant

105. A program checks whether every element in an array has appeared before using a hash set. What is the expected time complexity?

A. O(n)
B. O(log n)
C. $O(n^2)$
D. O(2^n)

106. A developer uses integer division where floating-point division was required. Which class of bug is most directly involved?

A. Incorrect arithmetic semantics
B. Memory leak
C. Race condition
D. Deadlock

107. A loop condition is `while (i <= n)` but i is never changed inside the loop. What is the likely issue when the condition initially holds?

A. Infinite loop
B. Stack overflow from recursion
C. Integer sorting
D. Successful termination

108. A program compares strings using reference identity when it needs content equality. What type of logical error can result?

A. Equal-looking strings may be treated as different objects.
B. The strings are always sorted automatically.
C. The program gains thread safety.
D. The comparison becomes O(1) in every language.

109. A function swaps adjacent elements repeatedly until no inversion remains. Which algorithmic idea does this resemble?

A. Bubble sort

B. Binary search

C. Hashing

D. Breadth-first search

110. A sorted array is searched by repeatedly comparing the target with the middle element and discarding half the range. What is the complexity?

A. O(log n)

B. O(n)

C. $O(n^2)$

D. O(2^n)

111. A program uses `x % 2` to decide whether an integer is even. Which property is being tested?

A. Remainder after division by two

B. Integer square root

C. Bitwise complement

D. Floating-point precision

112. A function receives an array and a target and must return whether two different elements sum to the target. Which expected-time strategy is strongest?

A. Hash set of previously seen values

B. Compare every pair only

C. Sort repeatedly inside the loop

D. Generate all permutations

113. A developer nests a loop of n iterations inside another loop of n iterations. What is the worst-case time complexity if each iteration performs constant work?

A. $O(n^2)$

B. O(n)

C. O(log n)

D. O(2^n)

114. A program stores a recursive function's calls on the call stack. What resource grows with recursion depth?

A. Stack space

B. Network bandwidth

C. Database rows

D. Disk partitions

115. A function returns the maximum value in an unsorted array. What lower-bound reasoning applies to a comparison-based scan?

A. Every element must potentially be inspected, so O(n) time is necessary.

B. O(log n) is always sufficient.

C. O(1) is sufficient without assumptions.

D. $O(n^2)$ is required.

116. A program has two independent loops, one O(n) and another O(n). What is the combined asymptotic complexity?

A. O(n)

B. $O(n^2)$

C. O(log n)

D. O(2n) as a distinct asymptotic class

117. A loop performs `i = i * 2` and then `i = i + 1` each iteration. Which statement is most accurate?

A. The multiplicative growth still gives logarithmic iteration count.
B. The loop is necessarily quadratic.
C. The loop is constant time for all n.
D. The increment makes it exponential.

118. A program copies an array into another array before modifying the copy. What property is being protected?

A. Independence of the copied data from later changes to the original
B. Network reliability
C. Database atomicity
D. Thread scheduling

119. A developer uses a floating-point number as a loop counter for equality with a target value. Why can this be risky?

A. Floating-point representation can make exact equality unreliable.
B. Floating-point values cannot be compared.
C. Floating-point arithmetic is always exact.
D. Loop counters must always be strings.

120. A function's output depends on an uninitialized variable. What kind of defect is this?

A. Undefined or indeterminate program behavior
B. Correct memoization
C. Stable deterministic computation
D. Normalization

121. A program reads n values and updates a single running minimum without storing the whole array. What is the auxiliary space?

A. O(1)
B. O(n)
C. $O(n^2)$
D. O(log n)

122. A recursive binary search is implemented incorrectly so the same search interval is passed again without shrinking. What is the likely consequence?

A. Non-termination or excessive recursion
B. Guaranteed O(log n) behavior
C. Automatic memoization
D. Constant-time completion

123. A program uses short-circuit AND and the left operand is false. What happens to the right operand in languages supporting short-circuit evaluation?

A. It may not be evaluated.
B. It must always be evaluated first.
C. It is executed twice.
D. It is converted into a loop.

124. A developer changes an algorithm from checking all pairs to sorting and using two pointers. What trade-off is typically being explored?

A. Lower time complexity at the possible cost of sorting and additional constraints
B. Guaranteed constant time
C. Elimination of all memory use
D. Removal of correctness requirements

125. A function repeatedly recomputes the same recursive subproblem. Which technique can reduce this repeated work?

A. Memoization

B. Randomization without state

C. Removing base cases

D. Increasing recursion depth

Data Structures & Algorithms

126. A browser's back-navigation needs to remove the most recently visited page first. Which data structure matches the access pattern?

A. Stack
B. Queue
C. Min-heap
D. Graph

127. A printer processes jobs in the order they arrive. Which structure naturally models this behavior?

A. Queue
B. Stack
C. Binary search tree
D. Hash set

128. A system repeatedly needs the smallest element and then inserts new elements. Which structure is designed for efficient minimum extraction?

A. Min-heap
B. Stack
C. Linked list
D. Hash table

129. A dictionary needs average constant-time key lookup. Which structure is the typical choice?

A. Hash table
B. Array with linear search
C. Stack
D. Queue

130. A singly linked list must insert a node immediately after a known node. What is the typical time complexity if the node reference is already available?

A. O(1)
B. O(log n)
C. O(n log n)
D. $O(n^2)$

131. A binary search tree becomes highly skewed because keys arrive in sorted order. What is the resulting worst-case search complexity?

A. O(n)
B. O(log n)
C. O(1)
D. O(n log n)

132. A balanced binary search tree keeps height proportional to log n. What benefit follows?

A. Search, insertion and deletion can remain O(log n) in the balanced case.
B. Every operation becomes O(1).
C. Traversal becomes O(log log n).
D. Sorting becomes unnecessary.

133. A graph problem asks for the shortest number of edges between two nodes in an unweighted graph. Which traversal is appropriate?

A. Breadth-first search
B. Depth-first search only
C. Heap sort
D. Binary search

134. A graph contains weighted edges that are all non-negative, and the task is single-source shortest paths. Which classic algorithm fits?

A. Dijkstra's algorithm

B. Binary search

C. KMP

D. Union-find alone

135. A graph has negative edge weights but no negative cycle. Which algorithm can handle single-source shortest paths?

A. Bellman-Ford

B. Binary search

C. Heapify

D. Counting sort

136. A minimum spanning tree is required for a connected weighted undirected graph. Which algorithms are appropriate?

A. Kruskal's or Prim's algorithm

B. BFS only

C. Binary search only

D. KMP only

137. Kruskal's algorithm needs to determine whether adding an edge would create a cycle. Which structure is commonly used?

A. Disjoint-set union

B. Stack only

C. Queue only

D. Trie

138. A prefix dictionary stores words so common prefixes share nodes. Which structure is most suitable?

A. Trie

B. Heap

C. Queue

D. Disjoint set

139. A dynamic programming solution stores the best answer for smaller subproblems and reuses them. What property makes this useful?

A. Overlapping subproblems

B. Random hashing

C. Packet retransmission

D. Referential transparency only

140. A greedy algorithm makes a locally optimal choice at each step. When is this strategy justified?

A. When the problem has a valid greedy-choice property and optimal substructure.

B. Whenever recursion is unavailable.

C. Whenever input is sorted.

D. For every optimization problem.

141. A two-pointer technique scans a sorted array from both ends to find a target sum. What is its typical time complexity?

A. O(n)

B. $O(n^2)$

C. O(log n) for the whole scan

D. O(2^n)

142. A sliding-window algorithm maintains a contiguous range while moving its boundaries. Which problem shape commonly benefits?

A. Subarray or substring constraints

B. Arbitrary graph coloring only

C. Tree balancing only

D. Matrix inversion only

143. A stable sorting algorithm preserves the relative order of equal keys. Why can this matter?

A. It preserves information encoded by prior ordering among equal-key records.

B. It guarantees O(1) space.

C. It always runs in O(n).

D. It prevents all comparisons.

144. Merge sort is applied to n elements. What is its typical time complexity?

A. O(n log n)

B. $O(n^2)$

C. O(log n)

D. O(2^n)

145. Quick sort chooses a poor pivot repeatedly on already ordered input. What worst-case behavior can result?

A. $O(n^2)$

B. O(log n)

C. O(1)

D. O(n) for every partition strategy

146. A dynamic programming solution uses a table indexed by position and remaining capacity. What is this an example of?

A. State-based tabulation

B. Hash collision resolution

C. Graph traversal

D. Stack unwinding

147. A recursive tree traversal visits left subtree, node, then right subtree. What traversal is this?

A. In-order

B. Pre-order

C. Post-order

D. Level-order

148. A topological ordering is required for tasks with prerequisite dependencies. What graph property is required?

A. The dependency graph must be a directed acyclic graph.

B. The graph must be complete.

C. The graph must be undirected.

D. All edges must have equal weights.

149. A cache uses least-recently-used eviction. Which pair of structures can support O(1) lookup and recency updates?

A. Hash map plus doubly linked list

B. Stack plus queue only

C. Binary search tree only

D. Array plus recursion only

150. A problem asks for the number of connected components in an undirected graph. Which approaches are appropriate?

A. DFS/BFS traversal or disjoint-set union

B. Binary search on vertices

C. Heap sort only

D. Trie traversal only

Software Engineering Fundamentals

151. A team divides a large application into components with clear interfaces so each can change internally without affecting callers. What principle is being pursued?

A. Encapsulation and modularity
B. Global coupling
C. Shared mutable state everywhere
D. Circular dependency

152. A class should expose only the operations clients need while hiding implementation details. Which OOP concept best fits?

A. Encapsulation
B. Inheritance
C. Recursion
D. Serialization

153. A subclass provides a specialized implementation of a method defined by its parent type. What concept is demonstrated?

A. Method overriding
B. Method overloading
C. Encapsulation
D. Aggregation

154. Two methods have the same name but different parameter lists within the same class. What is this commonly called?

A. Method overloading
B. Method overriding
C. Dependency injection
D. Serialization

155. A service receives its repository dependency from outside instead of constructing it internally. What design principle is illustrated?

A. Dependency injection
B. Tight coupling
C. Global state
D. Inheritance-only design

156. A module changes internally but its public contract remains stable. What property helps minimize impact on callers?

A. Abstraction through a stable interface
B. Direct access to every internal field
C. Global variables
D. Circular dependencies

157. A unit test fails after a refactor even though the implementation change was intended to preserve behavior. What should the team do first?

A. Determine whether the failure indicates a real regression or an outdated test expectation.
B. Delete the test immediately.
C. Ignore all failing tests.
D. Deploy before investigating.

158. A test verifies that a function returns the correct result for one isolated input without external services. What type of test is this?

A. Unit test
B. Load test
C. Penetration test
D. End-to-end deployment test

159. A test exercises the entire application through its public interface and real integrated components. What level is this?

A. End-to-end test
B. Unit test
C. Static type check
D. Code formatting

160. A bug occurs only when two threads update the same shared counter concurrently. What class of issue is likely?

A. Race condition
B. Syntax error
C. Memory leak only
D. Compilation warning

161. Two threads each hold a lock needed by the other and neither can proceed. What condition is this?

A. Deadlock
B. Cache miss
C. Serialization
D. Polymorphism

162. A code review finds that one function performs validation, persistence, email sending and report formatting. What design concern is most apparent?

A. Excessive responsibility and poor separation of concerns
B. Excessive encapsulation
C. Strong cohesion
D. Correct single responsibility

163. A database schema change must be applied consistently across environments and rolled back if deployment fails. What practice helps?

A. Versioned database migrations
B. Manual undocumented edits
C. Random schema changes
D. Disabling backups

164. A CI pipeline automatically builds and tests every pull request. What is the primary engineering benefit?

A. Early detection of integration and regression problems
B. Guaranteed bug-free production code
C. Elimination of code review
D. Removal of version control

165. A team uses Git branches and pull requests to integrate changes. What problem does version control primarily solve?

A. Tracking and coordinating changes to source code
B. Replacing runtime monitoring
C. Guaranteeing algorithmic optimality
D. Encrypting all application data

166. A deployment introduces a severe defect and the previous version is known to work. What operational capability is most useful?

A. Fast rollback to a known-good version
B. Deleting the repository
C. Increasing deployment batch size
D. Removing health checks

167. A service repeatedly retries a failed external request immediately and creates even more load during an outage. What improvement is appropriate?

A. Bounded retries with backoff and jitter

B. Infinite immediate retries

C. Removing all timeouts

D. Increasing request frequency

168. An API receives malformed JSON. Where should validation occur?

A. At the service boundary before business logic uses the data

B. Only after database insertion

C. Only in the UI

D. Never; clients are always trusted

169. A secret API key is committed to a public repository. What is the appropriate response?

A. Revoke/rotate the exposed secret and remove it from the secure configuration path.

B. Rename the variable and keep the same secret.

C. Hide the commit locally only.

D. Add the secret to another source file.

170. A system uses logs to diagnose production failures. What makes logs more useful?

A. Structured, contextual events with appropriate severity and correlation information

B. Logging every secret and password

C. Random messages without timestamps

D. Removing request identifiers

171. A service's response time increases as traffic grows because every request performs an expensive database scan. What should be investigated?

A. Query plans, indexes and access patterns

B. UI color schemes

C. Source-code comments only

D. Increasing log verbosity

172. A team wants to release a feature to a small percentage of users before full rollout. What technique fits?

A. Canary release

B. Big-bang release

C. Manual source deletion

D. Static compilation

173. A developer changes an implementation while keeping behavior unchanged and improves internal structure. What is this?

A. Refactoring

B. Feature expansion

C. Data migration

D. Incident response

174. A codebase has a dependency graph with cycles among modules. Why can this be problematic?

A. It increases coupling and can complicate testing, deployment and change isolation.

B. It guarantees modularity.

C. It guarantees faster builds.

D. It eliminates integration risks.

175. A service should fail fast when a downstream dependency is unavailable instead of consuming all worker threads waiting indefinitely. Which mechanism helps?

A. Timeout and circuit-breaker controls

B. Infinite blocking

C. Unlimited retries

D. Removing dependency health checks

Modern Engineering Awareness

176. A company wants virtual machines, networking and storage while retaining responsibility for the guest operating system. Which cloud model fits?

A. IaaS
B. PaaS
C. SaaS
D. FaaS only

177. A developer deploys application code while the cloud provider manages the runtime platform and underlying servers. Which model best fits?

A. PaaS
B. IaaS
C. SaaS
D. Colocation only

178. A user consumes a complete hosted business application through a browser without managing its servers. Which model is this?

A. SaaS
B. IaaS
C. Bare metal
D. Local-only deployment

179. A service automatically adds application instances when traffic increases and removes them when traffic falls. What capability is this?

A. Auto-scaling
B. Static linking
C. Normalization
D. Tokenization

180. A team deploys an application in containers so it runs consistently across environments. What is the main benefit?

A. Packaging the application with its dependencies in a portable unit
B. Eliminating all operating-system concerns
C. Guaranteeing zero vulnerabilities
D. Replacing source control

181. Containers on the same host generally share which underlying resource?

A. The host operating-system kernel
B. The host's user passwords
C. The same process namespace without isolation
D. The same application memory

182. A load balancer distributes requests across several healthy application instances. What primary goal does this support?

A. Traffic distribution and availability
B. Source-code compilation
C. Database normalization
D. Password hashing

183. A system replicates data across multiple nodes so one node can fail without losing the service's only copy. What property is being improved?

A. Fault tolerance
B. Tokenization
C. Serialization
D. Static typing

184. A cache stores frequently requested results close to the application. What is the primary performance benefit?

A. Reduced latency and fewer repeated expensive operations
B. Guaranteed data freshness
C. Elimination of authorization
D. Removal of persistence requirements

185. A cache serves stale customer data after the underlying database changes. What issue is involved?

A. Cache invalidation or freshness
B. DNS resolution
C. Compilation
D. Deadlock

186. A browser resolves `api.example.com` to an IP address before connecting. Which service performs this lookup?

A. DNS
B. SMTP
C. FTP
D. DHCP only

187. An HTTP client receives status code 404. What does the response generally indicate?

A. The requested resource was not found
B. The request succeeded and created a resource
C. The server has permanently moved every resource
D. Authentication definitely succeeded

188. An HTTP client receives status code 401. What does it generally indicate?

A. Authentication is required or has failed
B. The resource was permanently deleted
C. The server successfully created the resource
D. The request is a valid cache hit

189. A client sends a request to create a resource and the operation can safely be retried without creating duplicates when repeated. What HTTP property is relevant?

A. Idempotency
B. Compression
C. Fragmentation
D. Encryption

190. An API uses HTTPS. What security property is primarily added compared with plain HTTP?

A. Encrypted and authenticated transport via TLS
B. Automatic database transactions
C. Guaranteed application-level authorization
D. Automatic input validation

191. A web application uses a cookie containing an authentication session identifier. What must be considered?

A. Secure handling such as appropriate cookie flags, expiration and session protection
B. Cookies are automatically immune to theft
C. Authentication no longer requires server-side controls
D. HTTPS is unnecessary

192. A user enters SQL syntax into a form and the application concatenates it directly into a query. What vulnerability is likely?

A. SQL injection
B. DNS poisoning
C. Race condition
D. Buffer compression

193. A web page inserts untrusted user input into HTML without proper output encoding. What vulnerability may result?

A. Cross-site scripting
B. Deadlock
C. Cache stampede only
D. Packet fragmentation

194. A server stores user passwords as plaintext. What is the appropriate improvement?

A. Store salted password hashes using a password-hashing algorithm designed for credentials.
B. Encrypt passwords with a reversible key and log the key.
C. Store passwords in browser cookies.
D. Base64-encode the passwords.

195. A distributed service needs to process a burst of events asynchronously instead of forcing producers to wait for consumers. What component fits?

A. Message queue or event broker
B. DNS resolver
C. CSS stylesheet
D. Static linker

196. A message queue consumer crashes after receiving a message but before completing processing. What design concern becomes important?

A. Delivery semantics and idempotent processing
B. UI responsiveness only
C. DNS caching
D. Source formatting

197. A database transaction must ensure that all changes happen or none do. Which ACID property is this?

A. Atomicity
B. Consistency only
C. Isolation only
D. Durability only

198. Two concurrent transactions should not expose intermediate changes to each other. Which ACID property addresses this?

A. Isolation
B. Atomicity
C. Durability
D. Availability

199. A database must preserve committed data even after a crash. Which property is relevant?

A. Durability
B. Isolation
C. Tokenization
D. Idempotency

200. A service exposes metrics for request rate, error rate and latency. What engineering capability does this primarily support?

A. Observability and operational monitoring

B. Source-code formatting

C. Data normalization

D. Model pretraining